The definition of a quasi-category

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Abstract

Quasi-categories are simplicial sets which fill inner horns. First introduced by Boardman and Vogt in [BV73], they have recently become the standard model for $(\infty, 1)$-categories, largely due to the work of André Joyal ([Joy08a, Joy08b]) and Jacob Lurie ([Lur09, Lur25]). We will introduce the requisite theory of categories and simplicial sets to understand the definition of a quasi-category. We will then explore some of the basic definitions to better understand what an $(\infty, 1)$-category truly is.

Contents

1 Introduction

In this talk, we will introduce quasi-categories as models for ∞-categories. This talk is technically accessible to the advanced undergraduate student with some understanding of category theory, but preferably the reader member should have had exposure to some algebraic topology and simplicial homotopy.

A good reference for basic category theory is [ML98], and good resources to learn from are [Rie16, Lei14]. For a much more brief overview, the reader can recall my notes [Tal25a] from the last talk I gave to the math club. A good reference for the homotopy theory of simplicial sets is [GJ99], although we will not need the level of theory they develop. For more accessible and lower level introductions, I recommend [Rie11], [Fri23], or my own paper [Tal25b], which is currently in preparation.

2 Motivation

We recall from basic category theory that there is a notion of a morphism of categories given by a *functor*. We can in fact construct a category $\mathcal{C}\mathrm{at}$ of *small* categories, where "small" is a technical term used to appease the set theorists that the reader can safely ignore. Objects of $\mathcal{C}\mathrm{at}$ are (small) categories, and morphisms are functors between them.

However, given functors $F, G : \mathcal{C} \to \mathcal{D}$, we also have a notion of a *natural transformation* $\alpha : F \Rightarrow G$, which is a *morphism of functors*.

To summarize the situation, we have

- categories (objects),
- functors (morphisms of objects),
- and *natural transformations* (morphisms of morphisms).

In a sense, we have "two levels" of morphisms: a first level (functors) between our objects, and a second level (natural transformations) between our first level of morphisms. We call our first level of morphisms *1-morphisms* and our second level *2-morphisms*. A category with two levels of morphisms is called an *2-category*.

Definition 2.1. A (strict) 2-category $\mathcal{C}$ is a class of objects $\mathrm{Obj}(\mathcal{C})$, a class of 1-morphisms $1\text{-}\mathrm{Mor}(\mathcal{C})$ between objects, and a class of 2-morphisms $2\text{-}\mathrm{Mor}(\mathcal{C})$ between 1-morphisms, satisfying analogues of the category axioms.

We don't make precise the definition of a 2-category because it is *extremely long*. One has to spell out all of the regular category axioms, but then has to write them all out again for the 2-morphisms. Additionally, there are *two ways to compose 2-morphisms*, which adds even more axioms. We will not need to understand the two different compositions, so we recommend the reader see [ML98] or [Rie16] for an explanation. For now, we simply ask that the reader understand that the definition is quite lengthy and complex.

A natural question to ask next is whether there is a notion of a 3-category, which has 3-morphisms between 2-morphisms. Indeed such a notion exists, but it is here that we run into one of the main problems of higher category theory: *strong vs weak n-categories*. One would expect that in a 3-category, you can compose 1-morphisms just like you could in regular 1-categories, and that this composition is associative, and all the other axioms are satisfied. However, this is *not the* *type of 3-category we see in practice*. Such a 3-category is called an *strong* 3-category. However, in practice we see *weak* 3-categories, which are (roughly) defined as follows:

A weak 3-category has 3 levels of morphisms, where all the strict equalities one would expect for strong 3-categories are replaced with *higher isomorphisms*. To illustrate what we mean, we take the example of 1-morphisms. If we have a composable triple (h, g, f), then composition in a weak 3-category is *not* associative in general, meaning $(hg)f \neq h(gf)$ in general. However, there is a specified 2-isomorphism α called the (1-)*associator* such that $\alpha : (hg)f \cong h(gf)$. By replacing the strict equality condition with a higher isomorphism, we have weakkened the requirements of a 3-category.

We did not have to bother with strong vs weak 2-categories since the theories turn out to actually be equivalent, but this is not the case for 3-categories, nor is it the case for n-categories with $n > 2$. This becomes an issue because the definition of n-categories starts becoming extremely complex to write out fully when $n = 3$, and by $n = 4$ it is considered "impossible to work with". Even worse, nearly all of the other simple ways to define n-categories define *strong* n-categories, which are the wrong type, because we simply never see strong n-categories in nature.

Mathematicians have managed to define weak n-categories, but all of the definitions are extmrely complex. We will explore what is arguably the most useful one. To make this definition, mathematicians employ 4 tricks:

1. We ignore n-categories, and pass directly to weak ∞-categories. These are categories with n-morphisms for all n, meaning there is no highest level. We usually call these ω-categories, for reasons that we will discuss later.
2. We may also consider what are called (n, r)-categories. These are regular weak n-categories, with the following condition: whenever $k > r$, all k-morphisms are invertible.
3. However, both ω-categories and (n, r)-categories are very difficult to define for obvious reasons, so our third trick is to merge (1) and (2) and consider (∞, n)-categories. These are ω-categories where all morphisms above the nth level are invertible. A weak n-category can be viewed as an (∞, n)-category by taking all the morphisms above n to simply be the identity morphism, which doesn't really add any extra information. However, if we can't even define n-categories, how can we possibly hope to define ω-categories, even if we have this extra invertibility condition? The solution is trick 4:
4. We don't define these objects at all. Rather, we use a *model*. A model is a *different* mathematical object that we can view in a specific way that makes it *behave* in the way we *expect* an (∞, n)-category to be.

An example of (4) is as follows. An $(\infty, 0)$-category can be thought of as a topological space, by thinking of

- points as objects,
- paths as 1-morphisms,
- homotopies of paths as 2-morphisms,
- homotopies of homotopies as 3-morphisms,

and so on, with the usual identities and composition. Combined with a *straightening* procedure, this structure can indeed model $(\infty, 0)$-categories, which we call ∞-groupoids. In fact, Grothendieck's *Homotopy Hypothesis* states that the $(\infty, 1)$-category of ∞-groupoids is equivalent to the $(\infty, 1)$-category of spaces.

In this talk, we will disucss a model for $(\infty, 1)$-categories. $(\infty, 1)$-categories are *extremely* important, since they are the natural structure to encode a homotopy theory into. Given a category $\mathcal{C}$ with a model structure, we imagine taking the objects and 1-morphisms to be precisely those of $\mathcal{C}_{cf}$, and the 2-morphisms and higher are the *homotopies*. Due to their importance, we usually reserve the term "∞-category" to mean $(\infty, 1)$-category, and we refer to (∞, ∞)-categories by *ω-categories* due to the fact that ω is the first infinite ordinal number.

In this talk, we will develop some of the theory of simplicial sets, which is required to define quasi-categories, our model of choice for ∞-categories. Quasi-categories are one of the most powerful models for ∞-categories, and have been widely used throughout the literature for the past decade or two. We will then explore some of the basic theory of quasi-categories, trying to learn a little bit about what ∞-categories should *feel like*. Unfortunately, we won't have enough time to construct all the machinery required to show everything I would like to. However, we will be able to define and describe the following:

1. A class of objects, 1-morphisms, identities, domain, and codomain;
2. 2-morphisms (called homotopies), 2-identities, 2-domain, 2-codomain;
3. An (informal) schema for describing n-morphisms, and identities at every level. We leave the levels $n \geq 2$ informal because, as we explained earlier, explicit definitions at this level are difficult to write down to the point of being unuseable;
4. Composition of 1-morphisms, which is *well-defined and associative up to 2-morphism*. ∞-categories relax the conditions of weak categories even further: rather than explicitly stating 2-isomorphisms for every 1-morphism rule, and so on up the latter, we simplify the discussion by saying that even *composition* of 1-morhisms isn't well-defined, meaning two morphisms may have many composites. However, all composites are *homotopic* to each other, meaning there is a 2-(iso)morphism between them;
5. A homotopy category. We said ∞-categories model homotopy theories, and given a suitable homotopy theory on a category $\mathcal{C}$, we can always quotient by the homotopy relation, obtaining a homotopy category $\operatorname{Ho}\mathcal{C}$. We should be able to do the same for any ∞-category $\mathcal{C}$, which we will denote by $\mathrm{h}\mathcal{C}$;
6. ∞-functors between ∞-categories;
7. A way to view 1-categories as ∞-categories. Intuitively this corresponds to taking all higher homotopies to be trivial.

3 Simplicial Sets

Quasi-categories are built atop the theory of *simplicial sets*. We review the basic theory.

Definition 3.1. The simplex category $\mathbf{\Delta}$ is defined as

- Objects are finite nonempty ordinals $n + 1 = [n] = \{0, \ldots, n\}$;
- Morphisms are monotone functions, meaning $f : [n] \to [m]$ is a morphism if $x \leq y$ implies $f(x) \leq f(y)$.

Of course, we are only really concerned with two important types of morphisms in $\mathbf{\Delta}$, called the *cofaces* and *codegeneracies*.

Definition 3.2. Let $n \geq 1$. For each $i \in [n]$, define $d^i : [n-1] \to [n]$ as the function

$$d^i(x) = \begin{cases} x, & x < i \\ x+1, & x \geq i \end{cases}.$$

We call d^i the *ith coface*. Let $m \geq 0$. For each $i \in [m]$, define $s^i : [m+1] \to [m]$ as the function

$$s^i(x) = \begin{cases} x, & x \leq i \\ x-1, & x > i \end{cases}.$$

We call s^i the *ith codegeneracy*.

Intuition 3.3. Cofaces and codegeneracies have a very intuitive description. The ith coface $d^i : [n-1] \to [n]$ basically inserts $[n-1]$ into $[n]$ by *skipping* i. Everything below i gets mapped to itself, and anything equal or above i gets mapped up one spot, effectively skipping i. Similarly, codegeneracies $[n+1] \to [n]$ collapse $[n+1]$ into $[n]$ by *double counting* i. They map everything up to i to itself, and everything above i gets shifted down by 1. The effect of this is both i and $i+1$ map to i, effectively double counting i.

Cofaces and codegeneracies are important due to the following two theorems. It is a rite of passage to prove both of them, so I leave them as an exercise. For a sketch of the proof of the first, see [ML98].

Proposition 3.4 (Epi-Mono Factorization)**.** *Let $f : [n] \to [m]$ be a morphism in $\mathbf{\Delta}$. Then f has a* unique *representation as a composite of cofaces and codegeneracies:*

$$f = d^{i_1} \circ \cdots \circ d^{i_k} \circ s^{j_1} \circ \cdots \circ s^{j_h}.$$

This factorization is known as the epi-mono factorization.

Proposition 3.5 (Cosimplicial Identities)**.** *With domains and codomains as appropriate, the following identities hold.*

$$\begin{cases} d^j d^i = d^i d^{j-1}, & i < j \\ s^j d^i = d^i s^{j-1}, & i < j \\ s^j d^j = s^j d^{j+1} = 1 & \\ s^j d^i = d^{i-1} s^j, & i > j+1 \\ s^j s^i = s^i s^{j+1}, & i \leq j \end{cases}.$$

These are referred to as the cosimplicial identities.

We do not ask that the reader remember the cosimplicial identities, and we will reference them any time they are invoked. The reader will pick them up slowly as they work through the theory.

The category $\mathbf{\Delta}$ admits a geometric understanding. Define the standard n-simplex $|\Delta^n|$ as the n-dimensional equilaterial triangle. We call this a *geometric n-simplex.* You can assign an arbitrary ordering to the vertices of $|\Delta^n|$, and then assign a corresponding ordering to the *faces* of $|\Delta^n|$ by taking the ith face to be the face across from the ith vertex. The ith coface then corresponds to the ith *face* $d^i : \left|\Delta^{n-1}\right| \to |\Delta^n|$, which inserts $\left|\Delta^{n-1}\right|$ as the ith face of $|\Delta^n|$. Codegeneracies are more complex to explain, but luckily we will not need the geometric intuition of simplices, so we omit it.

We now move on to the theory of simplicial sets. These arose as powerful models for topological spaces. Classically, a simplicial set can be identified with a topological space through a process known as *geometric realization*, which takes a *massive* collection of geometric simplices, and glues them together according to some rules encoded in the simplicial set. This process is very powerful and deserves at least some mention, since the geometric realization is what we call a *Quillen equivalence*, meaning the homotopy theory of topological spaces is the *exact same* as the homotopy theory of simplicial sets. This is a powerful tool for homotopy theorists.

We will not be discussing the homotopy theory of simplicial sets at length, and will use them as a means to an end to introduce the notion of a quasi-category. This of course involves defining simplicial sets, which we now do.

Definition 3.6. A simplicial set X is a functor $X : \mathbf{\Delta}^{\mathrm{op}} \to \mathcal{S}\mathrm{et}$.

We often prefer not to think about simplicial sets as functors, and rather as collections of sets that have some interesting combinatorial properties. We will introduce this point of view momentarily, but we would like to point out some facts we get for free by viewing these objects as functors.

Notation 3.7. We write $\mathrm{s}\mathcal{S}\mathrm{et}$ for the category of simplicial sets, which is the functor category $\mathrm{Fun}(\mathbf{\Delta}^{\mathrm{op}}, \mathcal{S}\mathrm{et})$. There are a variety of other notations for this category, such as $\mathcal{S}\mathrm{imp}\mathcal{S}\mathrm{et}$, $\mathcal{S}\mathrm{et}_{\Delta}$, or $\mathcal{S}\mathcal{S}\mathrm{et}$. Given a simplicial set X, we write X_n for X evaluated at $[n]$. We write $d_i : X_n \to X_{n-1}$ for the image on the coface $X(d^i)$, and call d_i a *face operator*. Similarly, we write $s_i = X(s^i) : X_n \to X_{n+1}$ and call s_i a *degeneracy operator*. We typically do not write $X(\alpha)$ for more general morphisms α in $\mathbf{\Delta}$, and rather write α^* and specify that we do not mean precomposition. We call maps α^* *simplicial operators*.

Remark 3.8 (Simplicial Identities)**.** The cosimplicial identities (proposition 3.5) dualize to what we call the *simplicial identities*. By functoriality and contravariance, all we need to do is flip the order of composites. This means that face maps and degeneracy maps in a simplicial set X satisfy the *simplicial identities*:

$$\begin{cases} d_i d_j = d_{j-1} d_i, & i < j \\ d_i s_j = s_{j-1} d_i, & i < j \\ d_j s_j = d_j s_{j+1} = 1 & \\ d_i s_j = s_j d_{i-1}, & i > j+1 \\ s_i s_j = s_{j+1} s_i, & i \leq j \end{cases}.$$

Remark 3.9. Since $\mathrm{s}\mathcal{S}\mathrm{et}$ is a functor category, morphisms of simplicial sets are given by morphisms of functors, which are simply *natural transformations*. More explicitly, let $X, Y \in \mathrm{s}\mathcal{S}\mathrm{et}$. A morphism $f : X \to Y$ of simplicial sets is a natural transformation from X to Y. Recall that natural transformations f have a component f_x for each x in the domain of the relevant functors. Translating this into the language of simplicial sets, this means for each $n \geq 0$, there is a *function* $f_n : X_n \to Y_n$ such that for any i, the squares

$$\begin{tikzcd} X_n \arrow[r, "f_n"] \arrow[d, "d_i"'] & Y_n \arrow[d, "d_i"] \\ X_{n-1} \arrow[r, "f_{n-1}"'] & Y_{n-1} \end{tikzcd} \qquad \begin{tikzcd} X_n \arrow[r, "f_n"] \arrow[d, "s_i"'] & Y_n \arrow[d, "s_i"] \\ X_{n+1} \arrow[r, "f_{n+1}"'] & Y_{n+1} \end{tikzcd}$$

commute. The reason we only care about faces and degeneracies is because by proposition 3.4, *all morphisms in $\mathbf{\Delta}$ are composites of faces and degeneracies.* By functoriality of simplicial sets, we can take any simplicial operator and decompose it into degeneracies and faces by proposition 3.4. We then just glue together all the naturality squares for those faces and degeneracies, and get a naturality square for that operator.

Remark 3.10. The Yoneda lemma gives important simplicial sets. Given $[n] \in \mathbf{\Delta}$, the Yoneda embedding gives a simplicial set $\mathbf{\Delta}(-,[n]) : \mathbf{\Delta}^{\mathrm{op}} \to \mathcal{S}\mathrm{et}$. We denote $\mathbf{\Delta}(-,[n])$ by Δ^n, and call it the *standard n-simplex* or *standard n-simplicial set.* The geometric realization of the standard n-simplex is the standard geometric n-simplex. Also by the Yoneda lemma, we immediately know that faces d_j in Δ^n are given by *precomposition* by cofaces d^j. Similarly, degeneracies s_i are simply precomposition by codegeneracies s^i.

The proper way to think about simplicial sets is as follows. A simplicial set X is a collection of sets X_n for each $n \geq 0$, and a collection of faces and degeneracies satisfying the simplicial identities. Since all maps in $\mathbf{\Delta}$ factor uniquely as composites of cofaces and codegeneracies, and moreover they can be factored using *only the cosimplicial identities*, any collection of maps satisfying the simplicial identities will generate the proper operators for X to be functorial.

Remark 3.11. We now investigate the standard n-simplices, since these are highly important objects. Given a cosimplicial operator $f : [n] \to [m]$, the Yoneda embedding gives a morphism $f_* : \Delta^n \to \Delta^m$ by postcomposition. In particular, given a coface $d^i : [n-1] \to [n]$, there is a map $(d^i)_* : \Delta^{n-1} \to \Delta^n$. We also call this map a coface, and we also choose to denote it by $d^i : \Delta^{n-1} \to \Delta^n$, rather than standard postcomposition notation. This is because the Yoneda embedding gives what we call a *cosimplicial object* in the category $\mathrm{s}\mathcal{S}\mathrm{et}$.

Remark 3.12. By the Yoneda lemma, there is an isomorphism

$$\mathcal{S}\mathrm{et}^{\mathbf{\Delta}^{\mathrm{op}}}(\mathbf{\Delta}(-,[n]), K) \cong K[n]$$

natural in $[n]$. Recognizing that $\mathrm{s}\mathcal{S}\mathrm{et} := \mathcal{S}\mathrm{et}^{\mathbf{\Delta}^{\mathrm{op}}}$ and that $\Delta^n := \mathbf{\Delta}(-,[n])$, this is more conventionally written as

$$\mathrm{s}\mathcal{S}\mathrm{et}(\Delta^n, K) \cong K_n.$$

In other words, morphisms of simplicial sets $\Delta^n \to K$ correspond *exactly* to n-simplices of K. This is extremely important, because it means any n-simplex $\tau \in K_n$ completely determines a morphism $\sigma : \Delta^n \to K$, and any morphism $\sigma : \Delta^n \to K$ determines an n-simplex $\tau \in K_n$. Investigating the proof of the Yoneda lemma shows that τ is given by $\sigma(1_{[n]})$, and that the way τ determines σ is given by taking the epi-mono factorization (proposition 3.4) of each n-simplex, writing it as precomposing $1_{[n]}$ by cofaces and codegeneracies, then applying the fact that morphisms of simplicial sets commute with faces and degeneracies (remark 3.9). For this reason, morphisms $\sigma : \Delta^n \to K$ are sometimes called n-simplices, and we may even identify them with each other, writing σ for both the morphism and the n-simplex it identifies under Yoneda. Some authors take this further and define n-simplices of a simplicial set K as morphisms $\Delta^n \to K$, although we think this is too far and we simply abuse terminology as we see convenient.

We now introduce arguably the most important example of simplicial sets, the *nerve* of a category.

Definition 3.13. Let $\mathcal{C}$ be a category. We define the *nerve* $N(\mathcal{C})$ of $\mathcal{C}$ to be the simplicial set where

- n-simplices are given by functors $[n] \to \mathcal{C}$ where $[n]$ is viewed as a poset category. More explicitly, n-simplices are chains of $n+1$ objects and n composable morphisms between them:

$$c_0 \xrightarrow{f_1} c_1 \xrightarrow{f_2} \cdots \xrightarrow{f_{n-1}} c_{n-1} \xrightarrow{f_n} c_n.$$

- Degeneracies $s_i : N(\mathcal{C})_n \to N(\mathcal{C})_{n+1}$ insert the identity right after the ith morphism and insert c_i as the new $(i+1)$th object. More explicitly, s_i maps the n-simplex

$$c_0 \xrightarrow{f_1} \cdots \xrightarrow{f_i} c_i \xrightarrow{f_{i+1}} \cdots \xrightarrow{f_n} c_n$$

  to the $(n+1)$-simplex

$$c_0 \xrightarrow{f_1} \cdots \xrightarrow{f_i} c_i \xrightarrow{1_{c_i}} c_i \xrightarrow{f_{i+1}} \cdots \xrightarrow{f_n} c_n.$$

- Faces are split into two different types of faces, *inner faces* and *outer faces.*
  - Inner faces are faces $d_i : N(\mathcal{C})_n \to N(\mathcal{C})_{n-1}$ where $0 < i < n$, and are defined by composing the maps around the ith object. More explicitly, inner faces map the n-simplex

$$c_0 \xrightarrow{f_1} \cdots \xrightarrow{f_i} c_i \xrightarrow{f_{i+1}} \cdots \xrightarrow{f_n} c_n$$

    to the $(n-1)$-simplex

$$c_0 \xrightarrow{f_1} \cdots \xrightarrow{f_{i-1}} c_{i-1} \xrightarrow{f_{i+1} \circ f_i} c_{i+1} \xrightarrow{f_{i+2}} \cdots \xrightarrow{f_n} c_n.$$

  - Outer faces are faces $d_i : N(\mathcal{C})_n \to N(\mathcal{C})_{n-1}$ where $i = 0$ or $i = n$, and are defined by simply deleting the trailing morphism. More explicitly, given an n-simplex

$$c_0 \xrightarrow{f_1} c_1 \xrightarrow{f_2} \cdots \xrightarrow{f_{n-1}} c_{n-1} \xrightarrow{f_n} c_n,$$

    the 0th face is given by

$$c_1 \xrightarrow{f_2} \cdots \xrightarrow{f_{n-1}} c_{n-1} \xrightarrow{f_n} c_n$$

    and the nth face is given by

$$c_0 \xrightarrow{f_1} c_1 \xrightarrow{f_2} \cdots \xrightarrow{f_{n-1}} c_{n-1}.$$

4 Horns

Now that we understand some basic definitions, we would like to learn about *horns*. Horns are fundamental to the notion of a quasi-category, but they are nontrivial objects to understand. We begin with the notion of a simplicial subset.

Definition 4.1. Let X be a simplicial set. A *simplicial subset* of X is a simplicial set Y such that

- $Y_n \subseteq X_n$ for all n;
- The faces and degeneracies of Y agree with those of X:

$$d_i^X | Y_n = d_i^Y, \qquad s_i^X | Y_n = s_i^Y.$$

If Y is a simplicial subset of X, we write $Y \subseteq X$.

Note that definition 4.1 tells us what the faces and degeneracies of a simplicial subset must be. To construct a simplicial subset, we only need to specify the sets Y_n, and the subset condition gives us the faces and degeneracies for free.

Warning 4.2. Not all choices of subsets $Y_n \subseteq X_n$ yield a simplicial subset. For example, let $X_n = \mathbb{Z}$ for all n, and choose $Y_n = X_n$ for all $n \neq 1$, and $Y_1 = \varnothing$. There must be a degeneracy map $s_0 : Y_0 \to Y_1$, but there are no functions with an empty codomain. In particular, one must check the images of the faces and degeneracies lie within the sets Y_n.

Definition 4.3. Let $\alpha : X \to Y$ be a morphism of simplicial sets. Then there is a subsimplicial set $\operatorname{im} \alpha \subseteq Y$, called the *image of* α, with sets given by

$$(\operatorname{im} \alpha)_n = \operatorname{im}(\alpha_n : X_n \to Y_n),$$

where we define α_n by recalling the description of morphisms of simplicial sets given in remark 3.9.

Many important simplicial sets arise as simplicial subsets of the standard n-simplicial set Δ^n. The first one is the *face*.

Definition 4.4. Note by the Yoneda lemma that postcomposition by a coface gives a morphism of simplicial sets $(d^i)_* : \Delta^{n-1} \to \Delta^n$. Define the ith face of Δ^n, denoted by $\partial_i \Delta^n$, as the image $\operatorname{im}(d^i)_*$.

Remark 4.5. An important thing to note about the face is that since the epi-mono factorization in $\mathbf{\Delta}$ is unique, there is an isomorphism $\Delta^{n-1} \cong \partial_i \Delta^n$ given in one direction by postcomposing by d^i, and in the other by simply removing it. Under this isomorphism, the $(n-1)$-simplex $1_{[n-1]} \in \Delta^{n-1}_{n-1}$ corresponds to the $(n-1)$-simplex $d^i \in \partial_i \Delta^n$. By remark 3.12, morphisms $\partial_i \Delta^n \to X$ are precisely morphisms $\Delta^{n-1} \to X$, which are precisely $(n-1)$-simplices $\tau \in X_{n-1}$. Since morphisms $\Delta^{n-1} \to X$ are given by $1_{[n-1]} \mapsto \tau$, we have that morphisms $\partial_i \Delta^n$ are given by $d^i \mapsto \tau$.

Proposition 4.6. *Let* $\{X^i\}_{i \in I}$ *be a collection of simplicial sets* X^i *such that their faces and degeneracies all agree with each other on levelwise intersections:*

$$d^i_k | \bigcap_{i \in I} X^i_n = d^j_k | \bigcap_{i \in I} X^i_n$$

for all $i, j \in I$ *and* $d^i_k : X^i_n \to X^i_{n-1}$ *the* k*th face. Then the* levelwise union

$$\bigcup_{i \in I} X^i$$

is a simplicial set, and X^i *is a simplicial subset for all* i*.*

Proof. Indeed, by [ML98, §5.3 Thm. 1], colimits in functor categories such as $s\mathcal{S}et$ are computed levelwise. Since the union of sets is the pushout along inclusions of the intersection into component sets

$$\begin{tikzcd} X^i_n \cap X^j_n \arrow[r, hook] \arrow[d, hook] & X^i_n \arrow[d] \\ X^j_n \arrow[r] & X^i_n \cup X^j_n \end{tikzcd}$$

(and this generalizes for arbitrary unions), the statement is proven by cocompleteness of $\mathcal{S}et$. ■

The construction of the union of simplicial sets yields a simplicial set of fundamental importance: the *horn*.

Definition 4.7. Let $0 \leq i \leq n$. The ith horn Λ_i^n is defined as the simplicial subset

$$\Lambda_i^n := \bigcup_{j \in [n], j \neq i} \partial_j \Delta^n \subseteq \Delta^n.$$

Terminology 4.8. Let K be a simplicial set, and $\sigma_0 : \Lambda_i^n \to K$ a morphism of simplicial sets. We often refer to σ_0 as a *horn in* K. Similarly, we refer to a map $\partial_i \Delta^n \to K$ as a *face* of K.

Definition 4.9. Let K be a simplicial set, and let $\sigma_0 : \Lambda_i^n \to K$ be a horn in K. A *filler* for the horn σ_0 is a morphism of simplicial sets $\sigma : \Delta^n \to K$ which agrees when restricted along the horn $\sigma|\Lambda_i^n = \sigma_0$. Put another way, given the diagram of solid arrows

$$\begin{tikzcd} \Lambda_i^n \arrow[r, "\sigma_0"] \arrow[d, hook] & K \\ \Delta^n \arrow[ur, dotted, "\sigma"'] & \end{tikzcd}$$

a filler is a *not necessarily unique* dotted arrow as indicated, rendering the diagram commutative.

Horns and fillers are objects of particular interest. They play an important role in the homotopy theory of simplicial sets, but will also be of extreme importance in the theory of quasi-categories. The following theorem is one of the most important theorems regarding horns, as it gives us a convenient way to construct and study these objects.

Theorem 4.10. *Let K be a simplicial set and $0 \leq i \leq n$. A horn $\sigma : \Lambda_i^n \to K$ is fully determined by a collection of $(n-1)$-simplices $\tau_j \in K_{n-1}$ for each $j \neq i$, $0 \leq j \leq n$, satisfying the* compatibility condition*:*

$$\forall k < j,\ d_k(\tau_j) = d_{j-1}(\tau_k).$$

The horn σ is then defined by mapping $d^j \mapsto \tau_j$.

The proof is fairly lengthy so we unfortunately must omit it here. It is proven in [GJ99]. The theorem is important enough that we feel it is important to unpack its impliciations.

The theorem states that given a collection of simplices τ_j for $j \neq i$ satisfying the compatibility condition, the map Λ_i^n given by $d^j \mapsto \tau_j$ completely describes a horn $\sigma : \Lambda_i^n \to K$. This means that given this data, there is exactly one way to obtain a horn σ. This is obtained by applying the face and degeneracy operators to the cofaces d^j to describe all simplices of the ith horn Λ_i^n. The converse is also true, meaning given a horn there are maps $d^j \mapsto \tau_j$ satisfying the compatibility condition, but this is less interesting (and far easier to prove) than the other direction.

One of the upshots of theorem 4.10 is corollary 4.11:

Corollary 4.11. *Let $\sigma_0 : \Lambda_i^n \to K$ be a horn in a simplicial set K. With notation as in definition 4.9, a filler $\sigma : \Delta^n \to K$ is completely determined by an n-simplex $\tau \in K_n$ such that $d_j(\tau) = \tau_j$ for all $i \neq j$.*

Proof. By the Yoneda lemma (remark 3.12), a morphism $\sigma : \Delta^n \to K$ is equivalent to an n-simplex $\tau \in K$ such that $\sigma(1_{[n]}) = \tau$. For σ to be a filler simply means that $\sigma|\Lambda_i^n = \sigma_0$. By theorem 4.10, it suffices to show σ agrees with σ_0 on cofaces d^j with $j \neq i$.

To show this, note that $d^j = 1_{[n]} \circ d^j$, and thus

$$\sigma(d^j) = \sigma(1_{[n]} \circ d^j) = \sigma d_j(1_{[n]}),$$

where the last step follows by remark 3.10: faces in Δ^n are given by precomposition. By remark 3.9, morphisms of simplicial sets commute with faces, meaning

$$\sigma d_j(1_{[n]}) = d_j(\sigma(1_{[n]}) = d_j(\tau) = \tau_j,$$

where the last step follows by hypothesis. Therefore, σ is a filler for σ_0. ∎

5 Definition

We are finally prepared to define quasi-categories.

Definition 5.1. A *quasi-category* is a simplicial set $\mathcal{C}$ which fills *inner* horns. More explicitly, for all $0 < i < n$ and for all diagrams of solid arrows

$$\begin{tikzcd} \Lambda^n_i \arrow[r] \arrow[d, hook] & \mathcal{C} \\ \Delta^n \arrow[ur, dotted] & \end{tikzcd}$$

there exists a *not necessarily unique* dotted arrow as indicated, rendering the diagram commutative.

This definition was first introduced by Boardman and Vogt in [BV73] under the name *weak Kan complex*. Kan complexes are simplicial sets which fill *all* horns, not just inner horns. André Joyal expanded heavily on their work in [Joy08a, Joy08b] and developed much of the elementary theory of quasi-categories, showing that they were fantastic models for ∞-categories, as we will see. It was Joyal who introduced the term *quasi-category*, which is commonly used by the literature today. Jacob Lurie used quasi-categories as the foundational model for [Lur09], which is the standard reference on ∞-category theory today.

Quasi-categories are the standard ways to define ∞-categories. For the remainder of this talk, I want to show you what ∞-categories *feel like* using the language of quasi-categories. We will define objects, 1-morphisms, identities, composites, 2-morphisms, informally describe higher n-morphisms (recall these are hard to describe explicitly), ∞-functors of ∞-categories, and homotopy categories. I will also close with a few extra remarks that we will not have enough time to discuss the theory of. It will be helpful to begin by investigating the nerve $N(\mathcal{C})$ of a regular 1-category $\mathcal{C}$.

Proposition 5.2 ([Gro61]). *Let K be a simplicial set. Then the following statements are equivalent.*

1. *There is a category $\mathcal{C}$ and an isomorphism $K \cong N(\mathcal{C})$;*
2. *K admits* unique *fillers for inner horns, meaning for all $0 < i < n$ and all diagrams of solid arrows*

$$\begin{tikzcd} \Lambda^n_i \arrow[r] \arrow[d, hook] & K \\ \Delta^n \arrow[ur, dashed] & \end{tikzcd}$$

there is a unique *dashed arrow as indicated, rendering the diagram commutative.*

See [Lur09] Proposition 1.1.2.4 for a proof. Note that the objects of $\mathcal{C}$ are the vertices $N(\mathcal{C})_0$, and the morphisms of $\mathcal{C}$ are the edges $N(\mathcal{C})_1$. We also know that the faces $d_0, d_1 : N(\mathcal{C})_1 \to N(\mathcal{C})_0$ are defined by

$$d_0 = \mathrm{cod}, \qquad d_1 = \mathrm{dom}\,.$$

Finally, we know that the identity morphism 1_x on an object $x \in \mathcal{C}$ is simply the image $s_0(x)$ of the degeneracy $s_0 : N(\mathcal{C})_0 \to N(\mathcal{C})_1$. We might thnk that we can generalize these definitions to all quasi-categories, meaning for any quasi-category $\mathcal{C}$, we define

1. Objects are vertices $\mathcal{C}_0$;
2. Morphisms are edges $\mathcal{C}_1$;
3. Domain and codomain are the faces d_1, d_0, respectively;
4. Identity is the degeneracy s_0.

To see if these definitions work in general, we need to introduce a notion of *composition*. Since ∞-categories are supposed to be *weak*, composition should only hold up to 2-isomorphism (which we will call *homotopy*). The theory of quasi-categories takes a powerful approach of only having composition be well-defined up to homotopy. We will use this to motivate a definition of homotopy as well. Just as in 1-categories, we write $f : x \to y$ to denote that $f \in \mathcal{C}_1$ is an edge of a quasi-category with domain x and codomain y.

Let $f : x \to y$ and $g : y \to z$ in a quasi-category $\mathcal{C}$. Note that these define a *horn* $\sigma_0 : \Lambda^2_1 \to \mathcal{C}$: since $f, g \in \mathcal{C}_1$, we define $d^0 \mapsto g$ and $d^2 \mapsto f$; by theorem 4.10 it suffices to show the compatibility condition, which in this case states

$$y = d_0(f) = d_0\sigma_0(d^2) = d_1\sigma_0(d^0) = d_1(g) = y.$$

As we can see, this follows, giving the horn as claimed. Since $\mathcal{C}$ is a quasi-category, by definition 5.1 this fills to a 2-simplex $\sigma : \Delta^2 \to \mathcal{C}$ rendering the diagram

$$\begin{tikzcd} \Lambda^2_1 \arrow[r, "\sigma_0"] \arrow[d, hook] & \mathcal{C} \\ \Delta^2 \arrow[ur, "\sigma"'] & \end{tikzcd}$$

commutative. Define $\tau = \sigma(1_{[2]})$ as the image under the Yoneda isomorphism, as expressed in remark 3.12. We of course have $d_0(\tau) = g$ and $d_2(\tau) = f$. Define $h := d_1(\tau)$. We claim that h is a morphism $x \to z$. This follows from the simplicial identities of remark 3.8: recall that $d_i d_j = d_{j-1} d_i$ whenever $i < j$. It follows that

$$d_1 d_1(\tau) = d_{2-1} d_1(\tau) = d_1 d_2(\tau) = d_1(f) = x,$$

$$d_0 d_1(\tau) = d_0 d_0(\tau) = d_0(g) = z.$$

We define h as a *composite of g and f*. We write this as the diagram

$$\begin{tikzcd} & y \arrow[dr, "g"] & \\ x \arrow[ur, "f"] \arrow[rr, "h"'] & & z \end{tikzcd}$$

For this reason, given a 2-simplex $\tau \in \mathcal{C}_2$ with vertices x, y, z, we usually express the 2-simplex as a diagram

$$\begin{tikzcd} & y \arrow[dr, "d_0(\tau)"] & \\ x \arrow[ur, "d_2(\tau)"] \arrow[rr, "d_1(\tau)"'] & & z \end{tikzcd}$$

A more "geometric" way to think about it that *actually turns out to work* is as follows: given two composable morphisms $f : x \to y$ and $g : y \to z$, we can construct the diagram

$$\begin{tikzcd} & y \arrow[dr, "g"] & \\ x \arrow[ur, "f"] & & z \end{tikzcd}$$

This looks like a geometric horn. We label the top right face as the 0th face, the bottom face as the 1st face, and the top left face as the second face. By the quasi-category condition, this fills to an entire 2-simplex τ whose 1st face is h, and we write

$$\begin{tikzcd} & y \arrow[dr, "g"] \arrow[d, phantom, "\tau"] & \\ x \arrow[ur, "f"] \arrow[rr, "h"'] & {} & z \end{tikzcd}$$

Geometrically, the triangle is actually "filled", and we don't just have the morphisms around the boundary. The "filling" of the simplex is thought of as the diagram being "commutative", meaning h is the composite $g \circ f$. We say that the 2-simplex *witnesses* h as the composite of g and f. This terminology is useful.

Recall that the quasi-category condition *did not* require that the filler be unique. Since the 0th and 2nd faces are fixed for a given inner horn $\Lambda^2_1 \to \mathcal{C}$, the only thing that can change is the first face. In more conventional terminology, this means that two morphisms *do not need to have a unique composite.*

What we need to do now is to define a notion of homotopy, such that composites are unique up to homotopy. More precisely, we need to define a notion of homotopy, and then we need to show all the different composites are indeed homotopic.

Definition 5.3. Let $\mathcal{C}$ be a quasi-category, and let $f, g : x \to y$ be morphisms. A *homotopy* $\tau : f \to g$ is a 2-simplex $\tau \in \mathcal{C}_2$ such that

$$d_0(\tau) = 1_y, \qquad d_1(\tau) = g, \qquad d_2(\tau) = f.$$

Recall that earlier we wrote 2-simplices as "commutative" diagrams (the quotes are explained later). A homotopy $f \to g$ can thus be written as a diagram:

$$\begin{tikzcd} & y \arrow[dr, "1_y"] & \\ x \arrow[ur, "f"] \arrow[rr, "g"'] & & y \end{tikzcd}$$

To explain why this makes sense, we have to invoke some circular reasoning. Luckily, since we are only describing the *intuition* right now and not the actual math, circular reasoning is fine. Suppose we are able to show that with *some* definition of homotopy, all composites of f and g are homotopic. Suppose we are also able to show that f composed with 1_y is simply f. In regular 1-category theory, the diagram would simply say $f = g$, but this is ∞-category theory. The composite $1_y \circ f$ is *only well-defined up to homotopy*, meaning that there are other composites other than just f. In particular, the diagram tells us that g is one of these composites. We thus have two different composites for 1_y and f: f and g. Since composition is well-defined up to homotopy, we have that f and g are homotopic, as claimed. Thus, provided we can indeed show $1_y \circ f = f$ and that composition is well-defined up to homotopy, this definition is a good one.

This also explains why I put "commutative" in quotes. In higher category theory, since composites are only well-defined up to higher morphism, commutative diagrams must also include the additional datum of the higher morphisms that witness the relevant composites. Thus the diagram does not strictly commute, but commutes up to higher homotopy.

We now need to show that $1_y \circ f = f$, that homotopy is an equivalence relation, and that composites are all homotopic to each other. These will be the final results we prove in this paper.

Proposition 5.4. *Let $f : x \to y$ be a morphism in a quasi-category $\mathcal{C}$. Then there is a 2-simplex witnessing the composite $1_y \circ f = f$.*

Proof. As seen earlier, since $\operatorname{cod} f = \operatorname{dom} 1_y$, we obtain a horn $(f, -, 1_y) : \Lambda^2_1 \to \mathcal{C}$. Here I have adopted the useful notation $(f, -, 1_y)$ for horns, since in view of theorem 4.10 we may simply describe the horn by $f \mapsto d^0$ and $1_y \mapsto d^2$, with no map to d^1. Our notation is designed to reflect this.

Rather than simply claiming a filler exists since $\mathcal{C}$ is a quasi-category, we are interested in *constructing* a filler τ which has $d_1(\tau) = f$. We claim that $s_1(f) = \tau$ is such a filler. First, we must of course show it is indeed a filler. Note that by the simplicial identities of remark 3.8, we have $d_1 s_1 = d_2 s_1 = 1$, and thus

$$d_1 s_1(f) = f, \qquad d_2 s_1(f) = f.$$

It suffices to show $d_0 s_1(f) = 1_y$, and indeed by the simplicial identities of remark 3.8, we have $d_i s_j = s_{j-1} d_i$ for $i < j$, giving us

$$d_0 s_1(f) = s_0 d_0(f) = s_0(y) = 1_y.$$

By corollary 4.11, $s_1(f)$ is indeed a filler, and since $d_1 s_1(f) = f$, we have that $s_1(f)$ witnesses f as the composite $1_y \circ f$, concluding the proof. ■

Proposition 5.5. *Let $\mathcal{C}$ be a quasi-category, and write $\mathcal{C}_1(x, y)$ for the set of morphisms $x \to y$ in $\mathcal{C}$. For $f, g : x \to y$ in $\mathcal{C}$, write $f \sim g$ if there exists a homotopy $f \to g$. Then $\sim$ is an equivalence relation on $\mathcal{C}_1(x, y)$.*

Proof. By proposition 5.4, we have that $\sim$ is reflexive. Symmetry and transitivity will follow from an intermediary result, which we now show. Let $f, g, h : x \to y$, and let $\sigma : f \to g$ and $\tau : h \to g$ be homotopies (note the direction of τ). Define $\varphi := s_1 s_0(y) : \Delta^2 \to \mathcal{C}$ to be the constant map at y. Note here that we are abusing notation and writing φ for both the morphism and the 2-simplex. We will show there is a 3-simplex α filling the horn

$$(\varphi, -, \tau, \sigma) : \Lambda^3_1 \to \mathcal{C},$$

where we are using our new horn notation introduced in proposition 5.4. We first of course show this is a horn; by theorem 4.10 we must show the compatibility condition. Note that σ, τ are homotopies both with domain f:

$$d_2(\sigma) = d_2(\tau) = f, .$$

This shows the compatibility condition for σ and τ, so now we show it for σ and φ. By definition 5.3, the 0th face of any homotopy is the identity, so we have $d_0(\tau) = 1_y$. We also have

$$d_1(\varphi) = d_1 s_1 s_0(y) = s_0(y) = 1_y$$

by the simplicial identities of remark 3.8. Thus, φ and τ are indeed compatible, giving the horn as claimed. Since $\mathcal{C}$ is an ∞-category, the horn admits a filler $\alpha : \Delta^3 \to \mathcal{C}$. We also write α for the 3-simplex given by Yoneda (remark 3.12). We now show $d_1(\alpha)$ is a homotopy $g \to h$. Indeed, by the simplicial identities of remark 3.8,

$$d_0 d_1(\alpha) = d_0 d_0(\alpha) = d_0(\varphi) = d_0 s_1 s_0(y) = s_0 d_0 s_0(y) = s_0(y) = 1_y,$$

$$d_1 d_1(\alpha) = d_1 d_2(\alpha) = d_2(\tau) = h,$$

$$d_2 d_1(\alpha) = d_1 d_3(\alpha) = d_1(\sigma) = g.$$

We thus have by definition 5.3 that $d_1(\alpha) : h \to g$ is a homotopy.

We now conclude by the following remarks. Conside the special case with $f = h$. The above situation then reads as follows: if there are homotopies $f \to g$ and $f \to g$, then there is a homotopy $g \to f$. This is precisely symmetry. From this we derive transitivity as follows:

1. Let there be homotopies $f \to g$ and $g \to h$;
2. By symmetry, there is a homotopy $g \to f$;
3. Invoking the above result (note we are swapping the roles of f and g) gives a homotopy $f \to h$.

■

Proposition 5.6. *Let $f x \to y$ and $g : y \to z$ in a quasi-category $\mathcal{C}$, and let h, h' be composites of g and f, witnessed by the 2-simplices σ and τ, respectively. Then h is homotopic to h'.*

Proof. We first show that $(s_1(g), -, \tau, \sigma)$ is a horn $\Lambda^3_1 \to \mathcal{C}$. Indeed,

$$d_0(\tau) = g, \qquad d_1 s_1(g) = g$$

by definition 5.3 and remark 3.8, and thus $s_1(g)$ and τ are compatible. To show τ and σ are compatible, we have

$$d_2(\sigma) = f = d_2(\tau),$$

and by theorem 4.10 we indeed have a horn. Since $\mathcal{C}$ is a quasi-category, there exists a filler $\alpha \in \mathcal{C}_3$. We claim that $d_1(\alpha) =: \varphi$ is a homotopy $h' \to h$. Indeed, note that

$$d_0(\varphi) = d_0 d_1(\alpha) = d_0 d_0(\alpha) = d_0 s_1(g) = s_0 d_0(g) = s_0(z) = 1_z,$$

$$d_1(\varphi) = d_1 d_1(\alpha) = d_1 d_2(\alpha) = d_1(\tau) = h',$$

$$d_2(\varphi) = d_2 d_1(\alpha) = d_1 d_3(\alpha) = d_1(\sigma) = h.$$

Therefore, φ is a diagram

$$\begin{tikzcd}
& z \arrow[dr, equal] & \\
x \arrow[ur, "h'"] \arrow[rr, "h"'] & & z
\end{tikzcd}$$

which is precisely a homotopy $h' \to h$ as given by definition 5.3. ■

A stronger result than proposition 5.6 is in fact available: a morphism h' is a composite of g and f *if and only if* it is homotopic to some composite h of g and f. See [Lur25, 0041] for proof of the result.

To summarize what we have done so far: we have defined

1. objects as vertices;
2. morphisms as edges;
3. domain and codomain as faces;
4. identity as degenerates;
5. composition as fillers for horns $\Lambda_1^2 \to \mathcal{C}$ given by composable pairs;
6. homotopies $f \to g$ by 2-simplices witnessing the composite $1 \circ f = g$;

and we have shown that composites are well-defined up to homotopy. Moreover, this agrees with the nerve construction in ways described above, meaning we may regard 1-categories as quasi-categories via their nerve. We also have likely also figured out that n-simplices are a good model for n-morphisms. First note that 1-simplices are 1-morphisms, and 2-simplicies are homotopies, which are our conventional name for 2-morphisms. Also note that 2-simplices often "witness" the equality of various 1-morphisms, and similarly 3-simplices often "witness" the equality of various 2-simplices. We could likely thus define 3-simplices as 3-morphisms, and define a (convoluded) composition schema for homotopies. It should be clear that these definitions generalizes to n-morphisms, albeit they might be extremely difficult to write down.

Quasi-categories give us a way to avoid this problem using the notion of a *mapping space*. Given vertices $x, y \in \mathcal{C}$, it is possible to define a *new* simplicial set $\mathcal{C}(x, y)$, where morphisms $x \to y$ are vertices, homotopies are 1-simplices, and so on. This definition is nontrivial and will not be explored in this paper, but it is quite convenient to work with, and is the way we avoid having to write down composition laws for arbitrary n-morphisms.

We conclude by briefly constructing ∞-functors and homotopy categories. The former is simple. An ∞-functor should be a functor on objects and n-morphisms for all n, which respects identity and composition. Morphisms of simplicial sets satisfy all of these: given $f : \mathcal{C} \to \mathcal{D}$ a morphism of simplicial sets,

- there is a function $f_n : \mathcal{C}_n \to \mathcal{D}_n$ for all n, and
- f preserves simplicial operators, which were used to define all important constructions, meaning f will preserve all important constructions we have made so far.

Therfore, morphisms of simplicial sets are ∞-functors. We now need to simply define a homotopy category.

Recall that a homotopy category should "identify homotopic morphisms". Incredibly, we don't need to consider anything beyond 2-morphisms, since the higher homotopic data is encoded in the existence of the 2-morphisms themselves. We thus construct the homotopy category $\mathrm{h}\mathcal{C}$ of a quasi-category $\mathcal{C}$ as follows:

- Objects are vertices $\mathcal{C}_0$;
- Hom-sets $\mathrm{h}\mathcal{C}(x, y)$ are given by quotienting the set $\mathcal{C}_1(x, y)$ by the homotopy relation;
- Composition is given by composition in $\mathcal{C}$.

Of course, one must show that composition is well-defined, but we are unfortunately out of time to show this. However, this does indeed follow, giving a well-defined notion of a homotopy category for a quasi-category $\mathcal{C}$.

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